

JEE MAIN REVISION MASTER: GRAVITATION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 208 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Gravitation - Core Application	208 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Gravitation - Core Application (Very High Yield — 208 PYQs)

Core Principle: Key problem pattern in Gravitation.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A body of mass m is taken from the surface of earth to a height equal to twice the radius of earth (R_e) . The increase in potential e..." [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "A planet (P_1) is moving around the star of mass $2M$ in the orbit of radius R . Another planet (P_2) is moving around another..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "Net gravitational force at the center of a square is found to be F_1 when four particles having mass $M, 2M, 3M$ and $4M$ are placed at th..." [Ans: A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Evening Shift (Q#1)

A body of mass m is taken from the surface of earth to a height equal to twice the radius of earth (R_e). The increase in potential energy will be _____. (g is acceleration due to gravity at the surface of earth)

- (A) $(1/2)mgR_e$
- (B) $(3/4)mgR_e$
- (C) $(1/4)mgR_e$
- (D) $(2/3)mgR_e$ [Correct]

Correct Answer: D

Solution Analysis: The potential energy (U) of a mass m at a distance r from the centre of the Earth is: $U = -\frac{GM_e m}{r}$. As we know that $g = \frac{GM_e}{R_e^2}$, so $GM_e = gR_e^2$. $U = -\frac{mgR_e^2}{r}$. At the surface of the Earth, the distance from the centre is $r_1 = R_e$. So, the initial gravitational potential energy of body is;
 $U_i = -\frac{mgR_e^2}{R_e} = -mgR_e$. The body is taken to a height $h = 2R_e$ above the surface. The distance from the centre is $r_2 = R_e + h = R_e + 2R_e = 3R_e$. $U_f = -\frac{mgR_e^2}{3R_e} = -\frac{1}{3}mgR_e$. So, the increase in p